Christos-Grigorios Gkovaris

Final-year CSE Student | Software Engineering | Full-Stack Web Development

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Summary

I am a Computer Science and Engineering student with strong software development and robotics skills, experienced in full-stack development, machine learning, and ROS-based robotic simulations. I have assisted in electronics labs and volunteered as a Python instructor, developing both technical expertise and the ability to communicate complex concepts effectively. I am motivated to apply software engineering principles to large-scale scientific research and contribute to multidisciplinary teams.

Education

University of Ioannina — MEng, Integrated Master's Degree (5-year Engineering Program)
Computer Science and Engineering — **Ioannina, Epirus, Greece • October 2021 - September 2026**

https://www.cse.uoi.gr/

Experience

University of Ioannina — Ioannina, Epirus, Greece
Laboratory Assistant at Electronics — **Feb.2025-Jun.2025 & Feb.2026-Jun.2026**
Guided students in circuit design and troubleshooting using OrCAD, enabling 20+ students to successfully complete complex lab projects and improving lab workflow efficiency by reducing errors.

Skills

Robotics & Simulation
ROS, Gazebo, RViz, trajectory planning

Programming & Backend
Python (pandas, NumPy, Scikit-learn, PyTorch basics), Java (Spring Boot, REST APIs), JavaScript (jQuery, AJAX)

Tools & Technologies
Git & GitHub, Linux (Ubuntu), Windows, Zorin OS, VMWare Player 17/17 Pro, VS Code, Eclipse IDE, Visual Studio 2022, VHDL & OrCAD (academic exposure)

Frontend
HTML5, CSS3, JavaScript Responsive web applications

Databases
MySQL, SQLite, MariaDB

Projects

(Master Thesis Project) Smart Wishlist Prioritization Flow - Chrome Extension for Intent Based Search Ranking — **February 2026 - September 2026**
(In progress . . .)

Neural Classification and Unsupervised Learning - Ground up implementation of supervised and unsupervised learning algorithms — **October 2025 - January 2026**
Developed a 3-layer Multi-Layer Perceptron (MLP) trained with backpropagation for supervised data classification, achieving accurate predictions on complex datasets. Implemented a K-Means clustering algorithm from scratch for unsupervised data analysis, engineering all underlying mathematical models and optimizing performance. Technologies: Java

SpringBoot Traineeship Management System - Web application for managing traineeships and internships — **February 2025 - June 2025**
Developed secure authentication and implemented instructor/trainee role separation, ensuring reliable access control and system security. Designed and optimized REST API endpoints for efficient data storage and retrieval, improving application responsiveness. Integrated data management logic following MVC architecture principles, enhancing maintainability and scalability of the web platform. Technologies: Java, Spring Boot, REST APIs, SQL

MLP Classifier for MNIST Digit Recognition - Python based neural network for handwritten number identification — **October 2024 - January 2025**
Implemented a Multi-Layer Perceptron classifier for accurate handwritten digit recognition, achieving high classification performance on the MNIST dataset. Engineered robust data preprocessing pipelines and optimized model hyperparameters for improved convergence and reliability. Evaluated results using confusion matrices and loss curves to ensure model accuracy and stability. Technologies: Python, Scikit-Learn, Pandas, Matplotlib

Treegram App Social Media Platform (Ruby on Rails) - Lightweight Instagram-inspired web application — **October 2024 - January 2025**
Implemented secure user authentication, photo uploads, and tagging, enabling a functional social media platform with controlled access. Developed a follow system to display posts from followed users in chronological order, enhancing user engagement. Built comment creation and deletion with user permissions, ensuring robust content management and interaction control. Technologies: Ruby on Rails, HAML, SQLite3, JavaScript/jQuery, AJAX

Jackal and RRbot Simulations - Real time robotic simulation and trajectory generation using ROS — **February 2023 - June 2023**
Developed motion planning and autonomous navigation for a differential drive robot and a 2-DOF robotic manipulator. Implemented trajectory generation using cubic polynomials and linear functions with parabolic blends in ROS, reducing simulated trajectory error by 15% and ensuring smooth real-time control. Engineered control nodes for real-time simulation and synchronized joint movement within a physics-based environment. Technologies: Python, ROS, Gazebo, RViz, Matplotlib, NumPy.

Certifications

Introduction to Linux — **November 2024**
LinkedIn

Firewall Administration Essential Training — **November 2024**
LinkedIn

Level Up: Python Data Acquisitions, Prep, and EDA — **November 2024**
LinkedIn

Volunteering

CSE Teacher (Python) - Secondary Education Tutoring Center GNOSI — **July 2024 - August 2024**
Chalkida, Evvoia, Greece
Delivered Python programming courses to 15–20 students, enabling hands-on completion of 5+ small projects and improving student engagement.

IT Support Specialist (Troubleshooting) - Secondary Education Tutoring Center GNOSI — **July 2023 - August 2023**
Chalkida, Evvoia, Greece
Provided comprehensive IT support and maintenance, resolving technical issues promptly and ensuring uninterrupted operations and optimal system performance across the organization.

Languages

Greek (Native)

English (Advanced)

French (Intermediate)

Swedish (Self-Study)